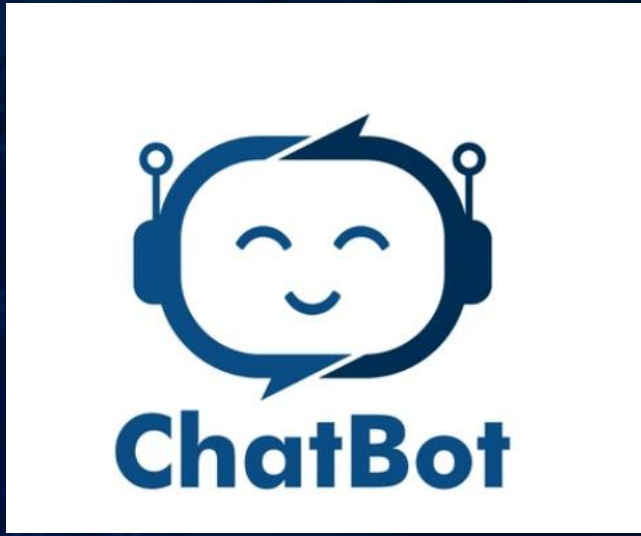
WELLBOT

AI POWERED WELLNESS & PRESCRIPTION ANALYSIS PLATFORM

Intelligent Healthcare Assistance with Hybrid AI Support
(MILESTONE 3) (TEAM A)



Problem Statement

- *Problems in Traditional Wellness Systems*
- No real-time AI support,
- Manual prescription understanding,
- No centralized admin monitoring,
- Limited personalization,
- Security risks.

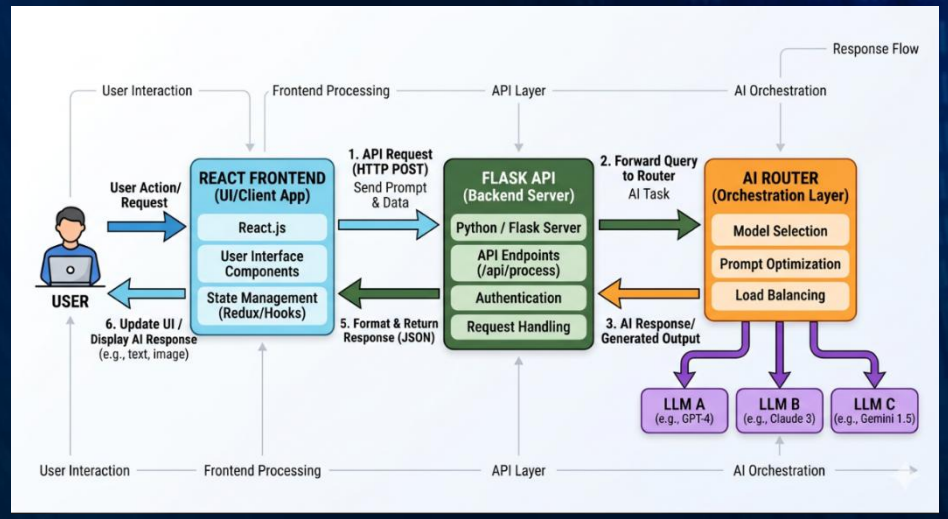
Proposed Solution

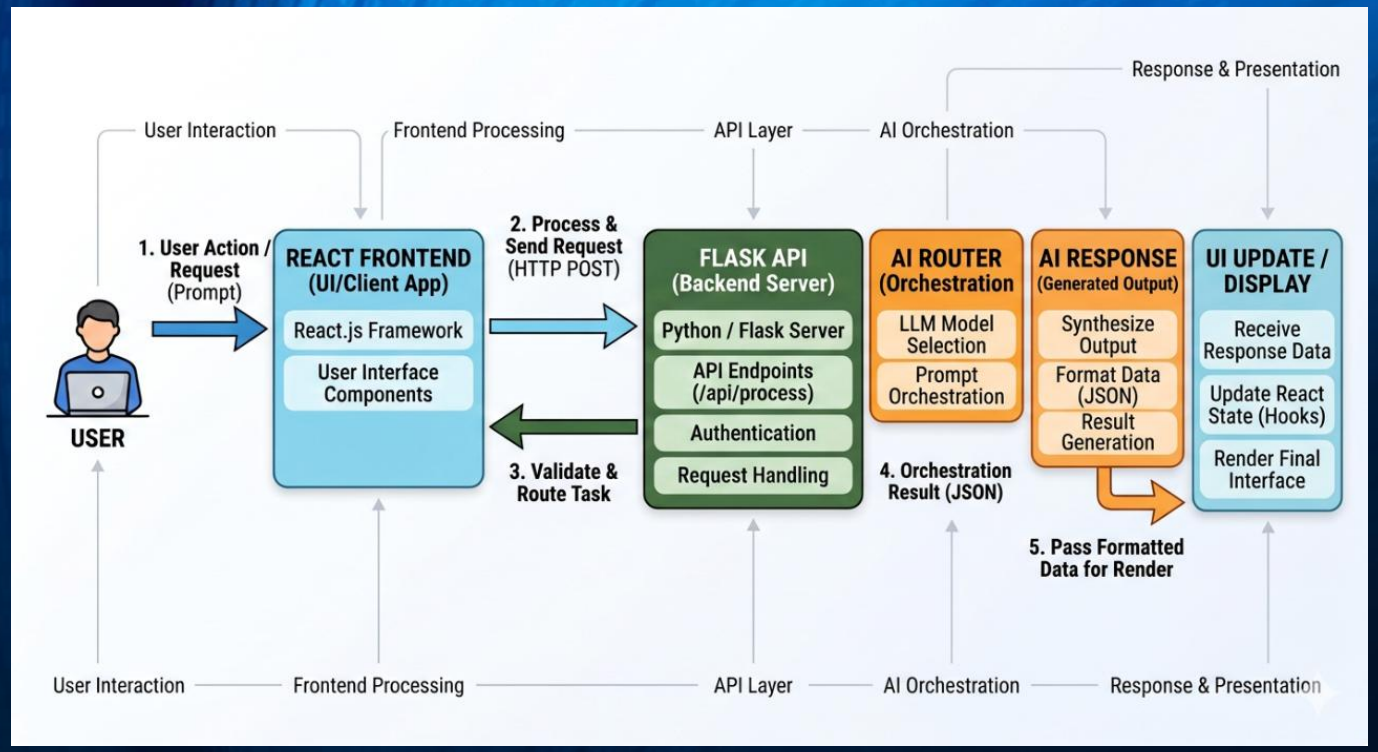
✅ *Our Solution: WellBot Platform*

- AI chatbot,
- Vision-powered analysis,
- Admin dashboard,
- Nutrition education,
- Hybrid AI.

❑ Real-time responses,
❑ Secure admin,
❑ Multi-LLM fallback.

System Architecture





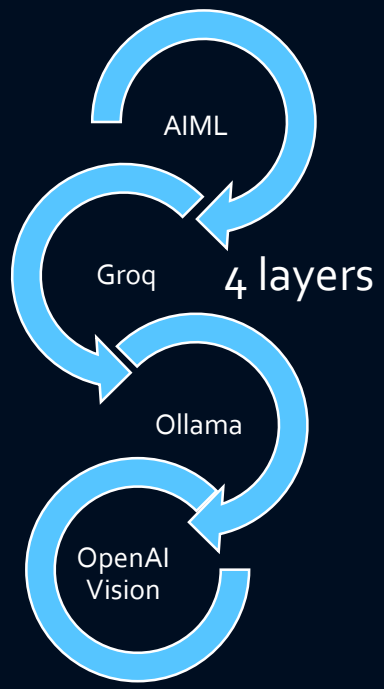
- **Frontend:** React UI, Sidebar Navigation, Image Upload
- **Backend:** Flask API, JWT Authentication, AI Model Router
- **AI Layer:** Groq (Llama 3.3), OpenAI GPT-4o-mini Vision, Ollama

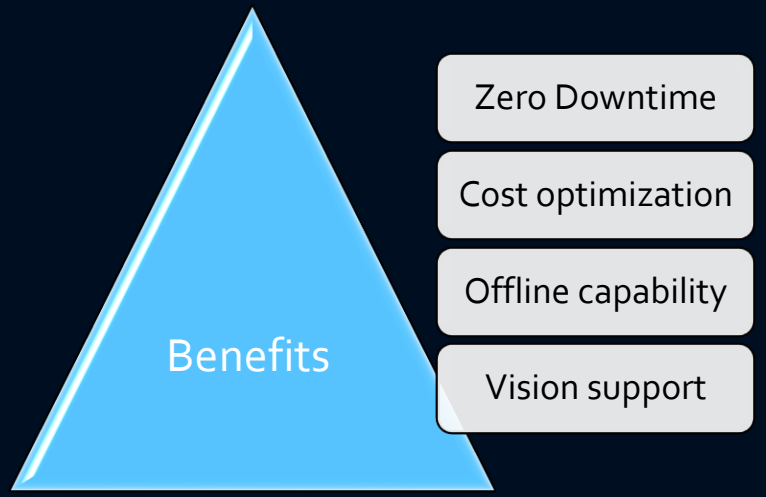
Flow:

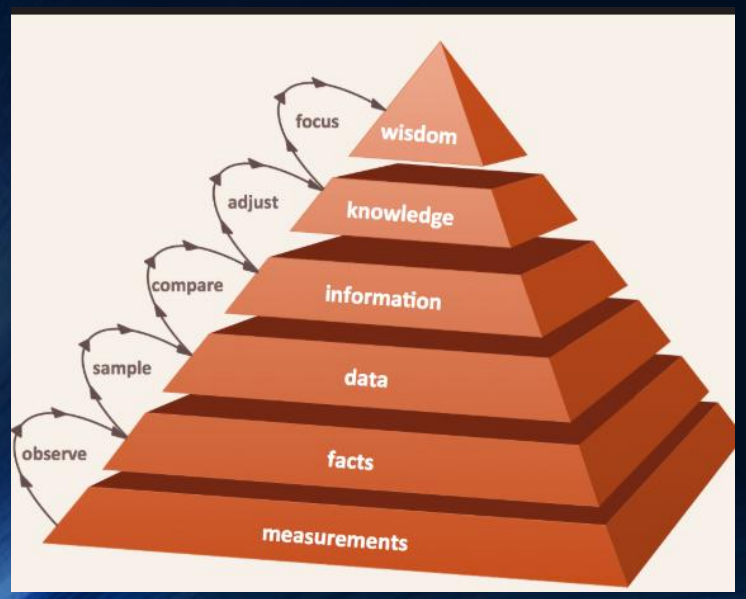
User → React → Flask → AI Router → Response → UI

Hybrid AI Model Integration

Intelligent AI Fallback System







✅ Benefits

Zero Downtime: Always responds, even if cloud models fail

Cost Optimization: Uses local models when possible

Offline Capability: AIML and Ollama work without internet

Vision Support: GPT-4o-mini handles image-based prescriptions

AI Vision & Prescription Analysis

AI Extracts:
Medicine names
Dosages
Instructions
Warnings

Benefits:
Reduces confusion
Helps elderly users
Quick clarity

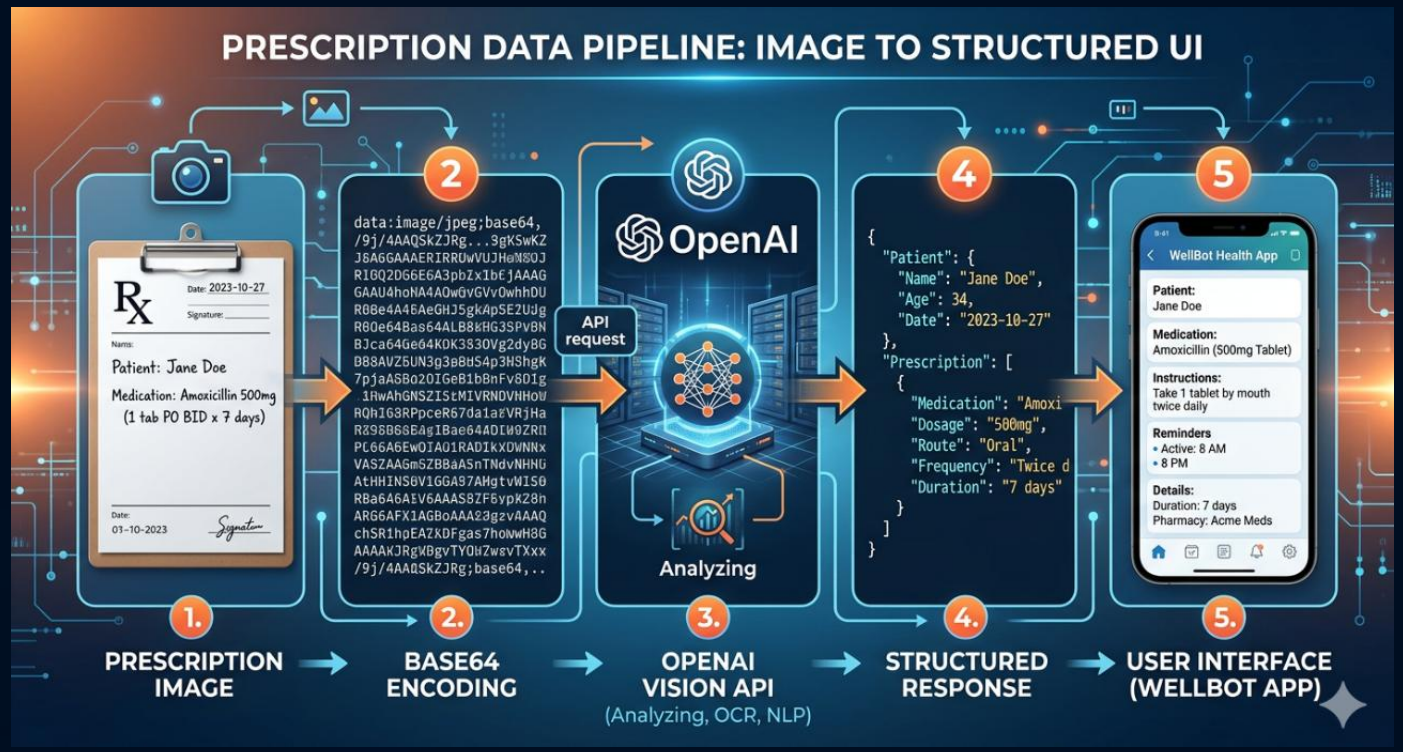


Image → Base64 → OpenAI API → Structured Response → UI

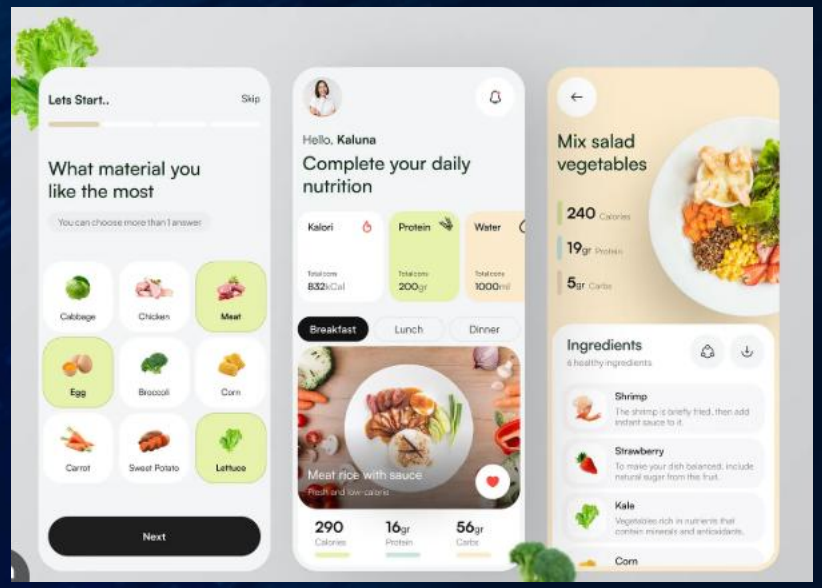
Admin Dashboard Expansion

•Features: User management, Chat monitoring, Issue reporting, Graphs.
•Analytics: Users, Queries, Feedback, Sessions.

User Dashboard & Nutrition Guide

- ***Personalized Wellness Experience***

•Sections: Overview, Nutrition Guide, Profile Settings.
•**Nutrition: Vitamins, Minerals, Benefits, Food sources.**



Security & Performance

- ***Enterprise-Level Security***

•Security: JWT, Admin protection, Session mgmt, API key safety.
•Performance: Async processing, AI daemon threads,
Auto-scroll UX, Lightweight React.

Future Enhancements & Conclusion

•Features: Voice chatbot, IoT, Multi-language, Cloud deployment.

•Conclusion: Scalable, Secure, AI-powered,
Vision-enabled,
Admin-controlled,
Hybrid LLM supported.

“Transforming Digital Wellness with Intelligent AI Systems”

THANK YOU

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